



Construction Quality Analytics Dashboard

Power BI Project inspired by QA/QC practices observed during my L&T Internship

Average Specific Gravity

2.58

Average Water Absorption (%)

1.64

Average Mechanical Test Value (%)

16.36

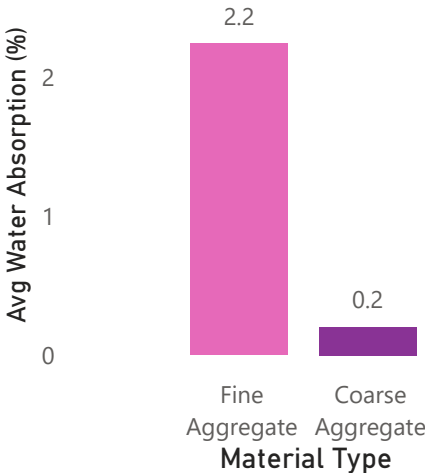
Average Bulk Density (g/cc)

1.49

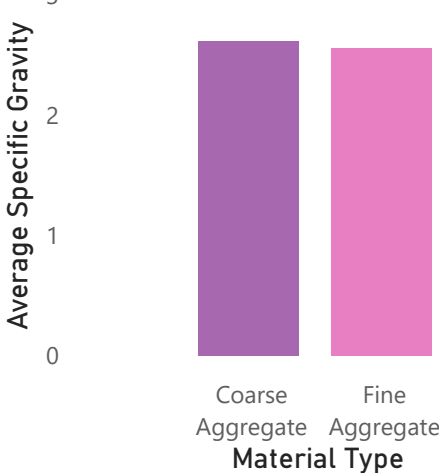
Maximum Bulk Density (g/cc)

1.54

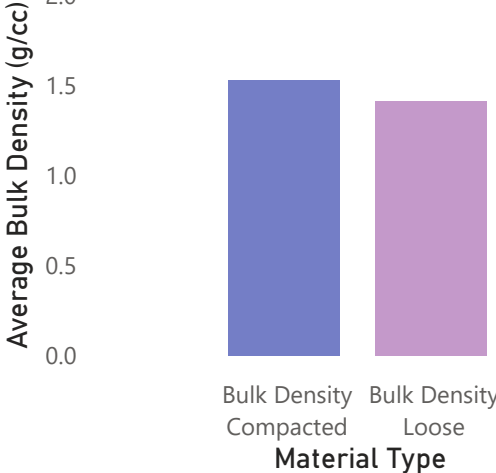
Average Water Absorption (%)



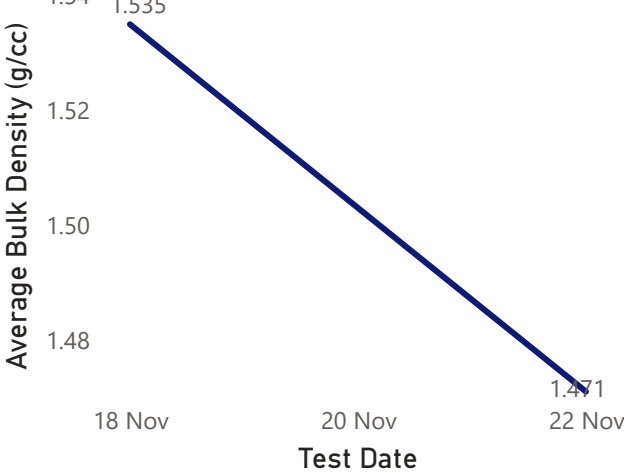
Average Specific Gravity



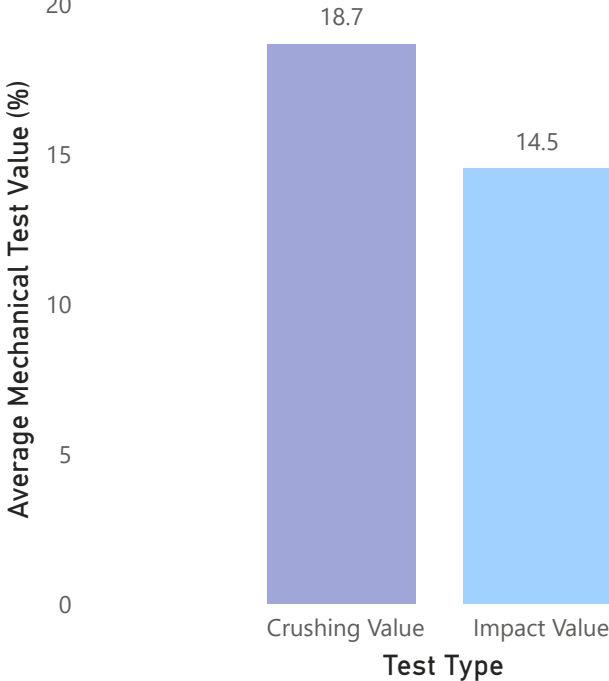
Average Bulk Density by Test Type (g/cc)



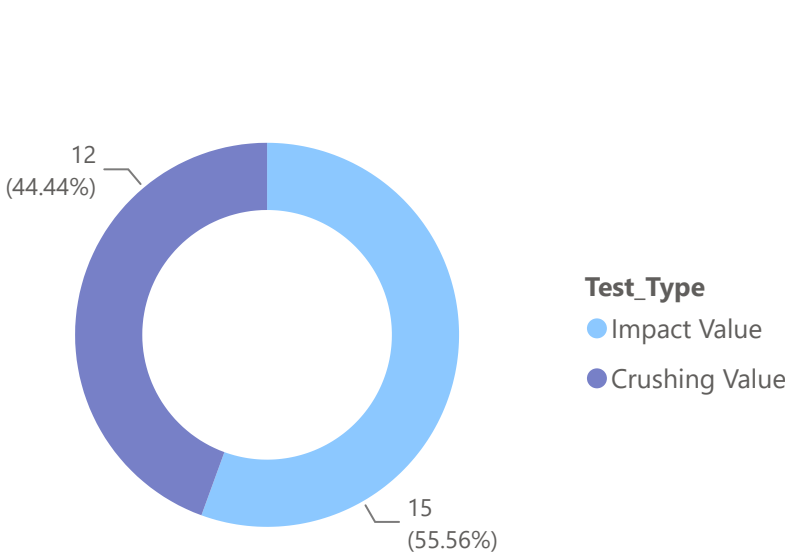
Bulk Density Trend (g/cc)



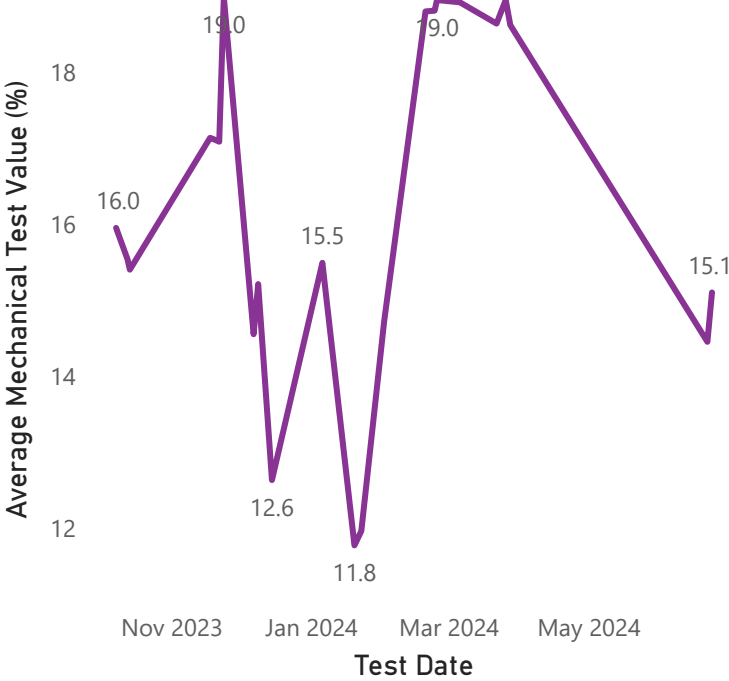
Average Test Value (%) by Test Type



Mechanical Test Distribution



Mechanical Test Trend (%)



QUALITY INSIGHTS

- Average Specific Gravity: **2.58**
- Average Water Absorption: **1.64%**
- Average Mechanical Test Value: **16.36%**
- Average Bulk Density: **1.49 g/cc**
- Maximum Bulk Density: **1.54 g/cc**
- Based on sample laboratory test data inspired by observations during my L&T internship at TIMS Hospital, LB Nagar.



QUALITY DASHBOARD – TIMS HOSPITAL

OVERVIEW

FILTERS



Year

- > ☐ 2023
> ☐ 2024
> ☐ 2025

Material Type

Coarse Aggregate

Fine Aggregate

Location



B/Plant

Batching Plant



AGGREGATE QUALITY ANALYTICS

Total Aggregate Tests

27

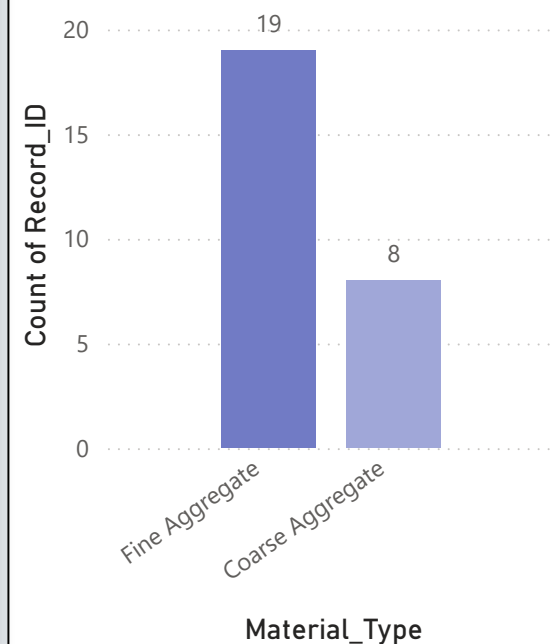
Average Specific Gravity

2.58

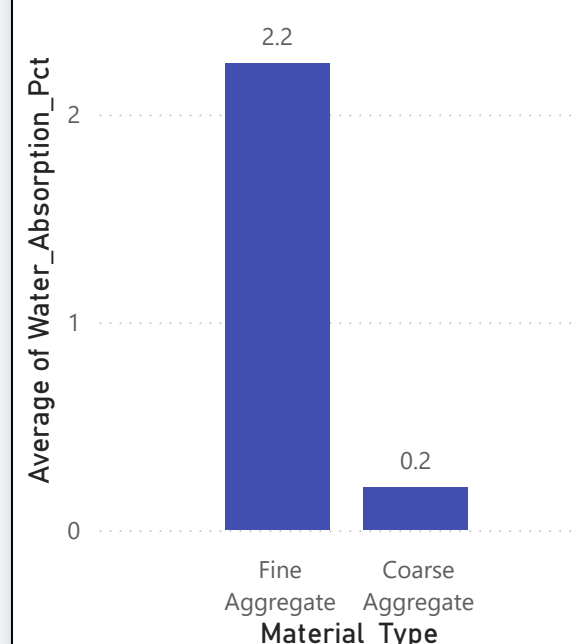
Average Water Absor...

1.64

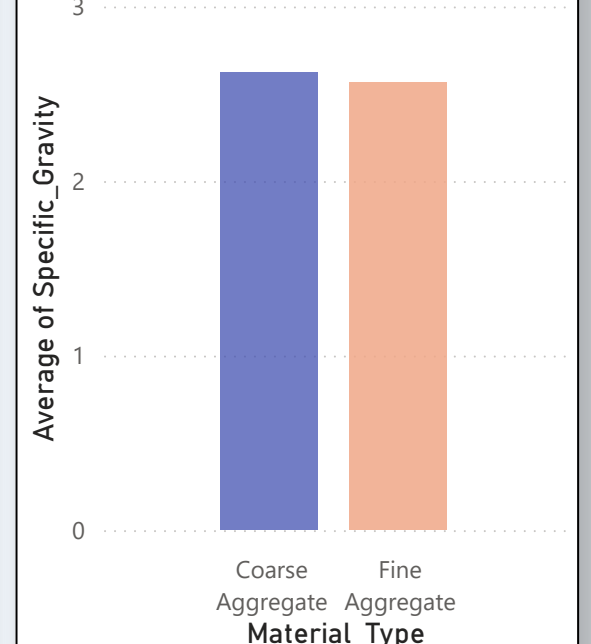
Test Count by Material



Average Water Absorption (%)



Average Specific Gravity



MECHANICAL TEST DASHBOARD

Total Mechanical Tests

27

Average Value

16.36

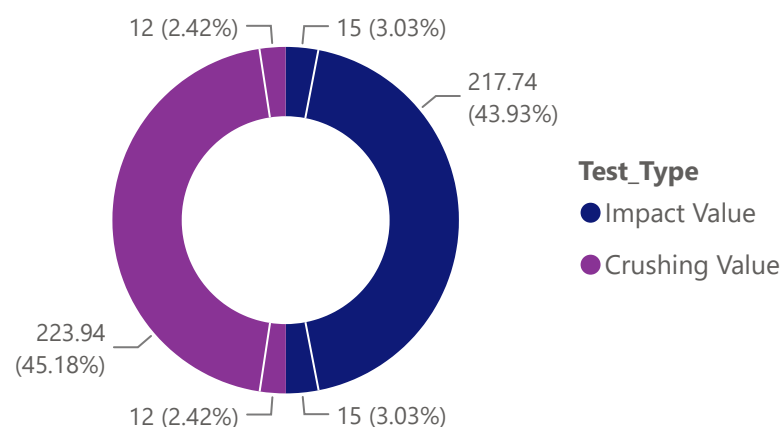
Maximum Test Value

18.99

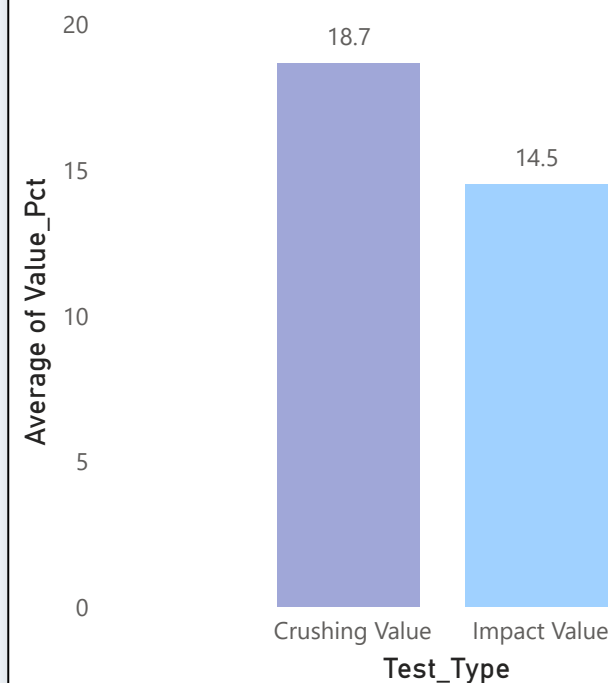
Minimum Test Value

11.77

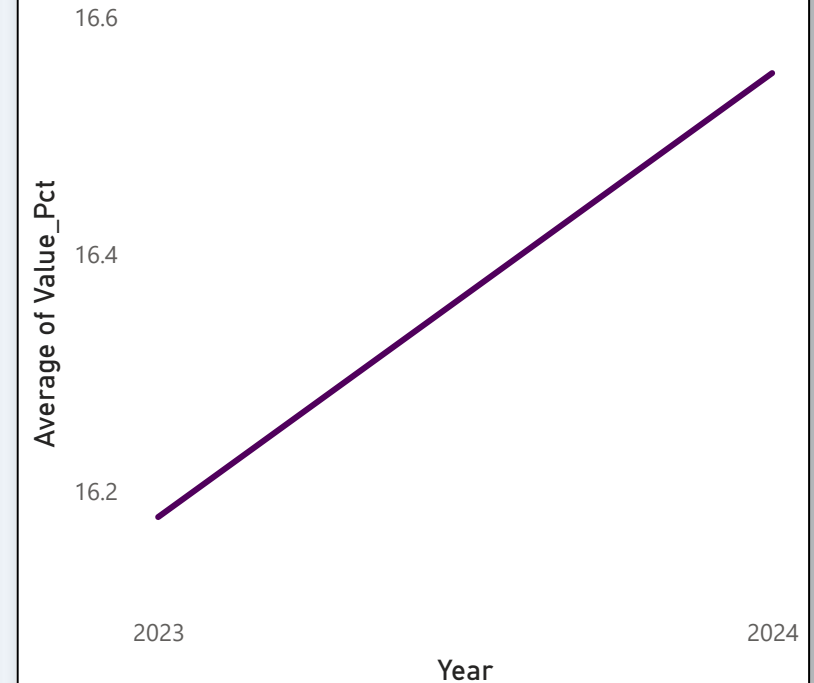
Test Type Distribution



Average Test Value (%) by Test Type



Average Test Value by Year



VOLUMETRIC TEST DASHBOARD

Total Volumetric Tests

11

Average Density

15.09

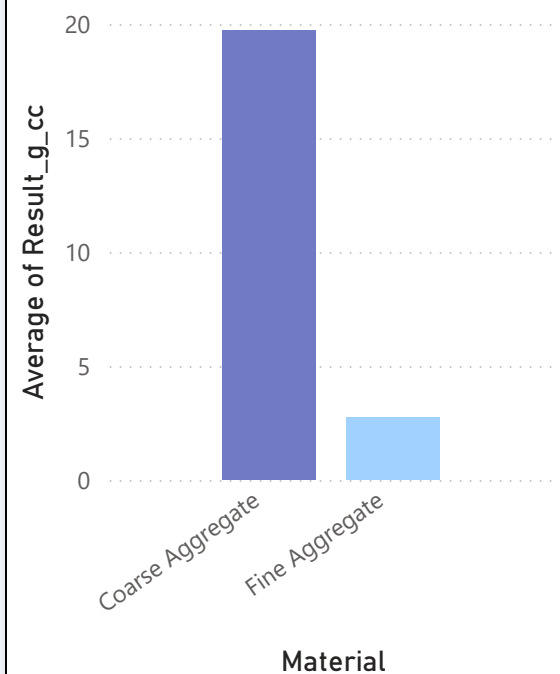
Maximum Density

34.02

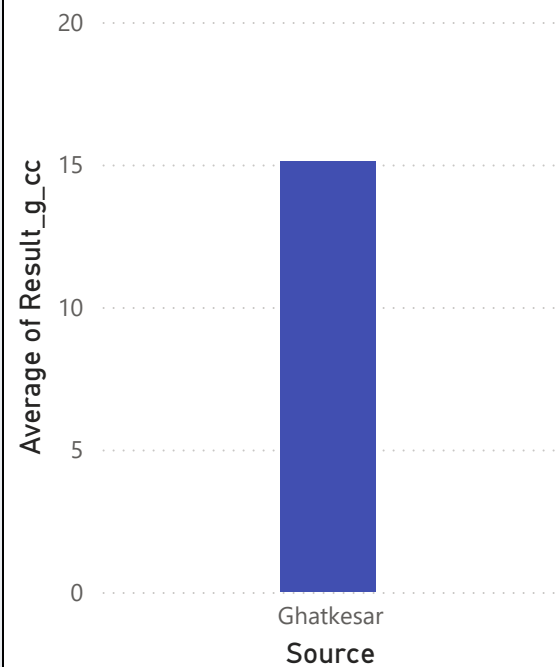
Minimum Density

1.41

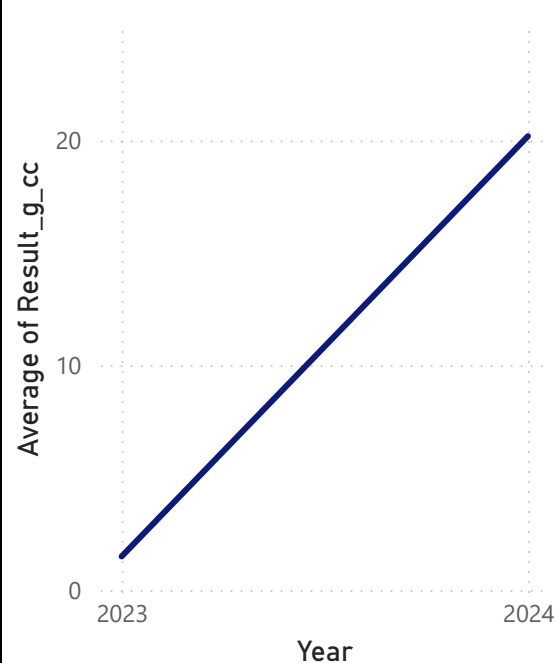
Density by Material



Density by Source



Density Trend



Density by Location

